2.5.3 Newmark Methods

As a special case of a partitioned Runge-Kutta method, consider the Newmark family of methods [280] defined for two parameters σ and η by the formulas

$$\boldsymbol{P} = \boldsymbol{p} - h(1-\sigma)\nabla U(\boldsymbol{q}) - h\sigma\nabla U(\boldsymbol{Q}),$$
$$\boldsymbol{Q} = \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{p} - h^2\left(\frac{1}{2}-\eta\right)\nabla U(\boldsymbol{q}) - h^2\eta\nabla U(\boldsymbol{Q}).$$

This is an exact transcription of the formulas in Newmark's original paper, only substituting $\boldsymbol{M}^{-1}\boldsymbol{p}$ for the velocity $\boldsymbol{v}$ wherever it appears. In practice the choice $\sigma = 1/2$ is used to avoid spurious damping (it can be demonstrated for a simple model problem); this certainly would appear to be desirable in the setting of molecular dynamics. For $\eta = 0$ we then arrive at the Verlet method. For other values of η the scheme is clearly implicit, which likely is the reason it is rarely used in molecular simulation, although it is popular in structural mechanics. The implicit Newmark methods are not symplectic, but a related family of symplectic methods can be constructed by replacing interpolated forces by forces evaluated at interpolated positions [395].

2.5.4 Multiderivative Methods

All of the methods discussed above rely on computing values of the vector field (i.e. momenta and forces) only. However, as we saw earlier in this chapter, in the setting of a Taylor series method, we may approximate a single step by

$$z_{n+1} = z_n + h\dot{z}_n + \frac{h^2}{2}\ddot{z}_n + \ldots + \frac{h^k}{k!}z_n^{(k)},$$

where it is possible to make use of higher order derivatives of the solution in formulating the method. Then using the differential equation, the time derivatives may be replaced by elementary differentials of the vector field. This same idea can be used in a more sophisticated way to improve the accuracy of molecular dynamics methods. Letting $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$, the Takahashi-Imada method [355] (also known as Rowlands' method [316]) has the same form as the Verlet method

$$\hat{\boldsymbol{P}} = \boldsymbol{p} - (h/2)\nabla\tilde{U}(\boldsymbol{q}),$$
$$\boldsymbol{Q} = \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}},$$
$$\boldsymbol{P} = \hat{\boldsymbol{P}} - (h/2)\nabla\tilde{U}(\boldsymbol{Q}),$$

where the corresponding potential energy function is

$$\tilde{U}(\boldsymbol{q}) = U(\boldsymbol{q}) - \frac{h^2}{24}\nabla U(\boldsymbol{q})^T \boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}).$$

The forces arising from such a modified potential can be worked out:

$$\tilde{\boldsymbol{F}} = -\nabla\tilde{U} = -\left[\boldsymbol{I} + \frac{h^2}{12}U''\boldsymbol{M}^{-1}\right]\nabla U,$$

where U'' is the Hessian matrix of the potential. It may in some cases be a daunting task to compute the Hessian matrix as would be required in the Takahashi-Imada method, but this is not always the case. Advantage can be made of the fact that for systems with short-ranged potentials, the Hessian matrix is likely to be very sparse, in which case this computation can be effected with little additional overhead, although this will depend on the underlying computer architecture, the size of the system, etc.

This method can be shown to have effective order four, meaning that there is a change of variables $\boldsymbol{\chi}_h$ which can be used to transform the Takahashi-Imada method into one of order four using the processing technique of Sect. 2.4.5. The potential energy modification has been specifically chosen to annihilate terms in the local error expansion (after coordinate transformation).

A discussion and comparison of several multiderivative methods for molecular applications may be found in [239].

2.5.5 *Other Methods*

We have mentioned previously that it is possible to reduce the Verlet method to a scheme involving positions only:

$$\boldsymbol{q}_{n+1} - 2\boldsymbol{q}_n + \boldsymbol{q}_{n-1} = h^2\boldsymbol{M}^{-1}\boldsymbol{F}(\boldsymbol{q}_n).$$

This is a type of a *multistep* method. Such methods may be studied using a generalization of the techniques used to understand one-step methods [167]. There are a variety of multistep methods which could in principle be used for molecular dynamics, however, we regard the benefits as unproven; in particular, such methods neglect the phase space structure such as the symplectic property.

Another scheme which is sometimes used in molecular dynamics is Beeman's Algorithm [30, 331].